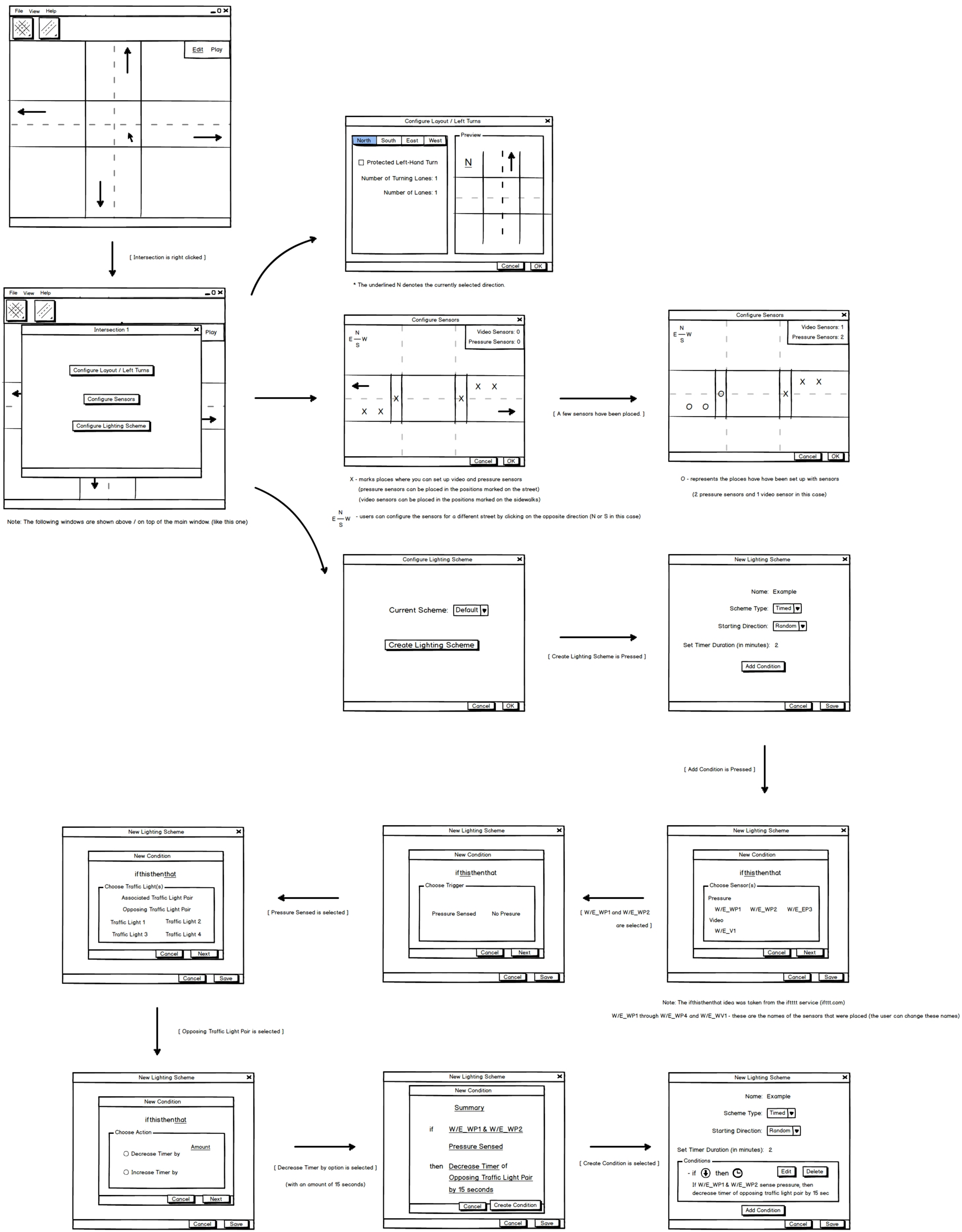
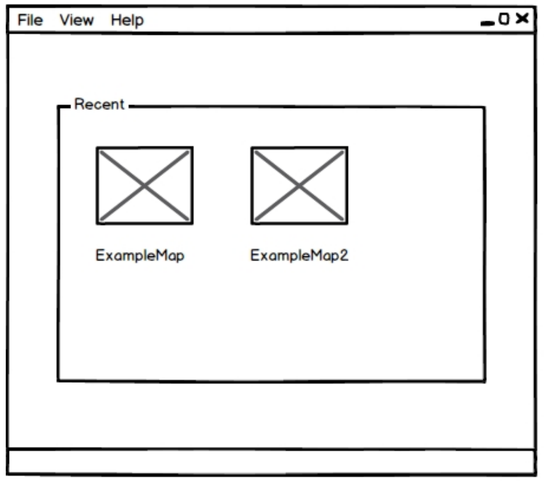
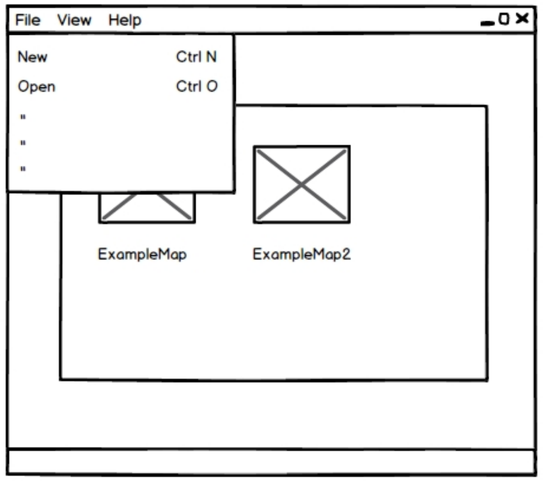
Configure Intersection



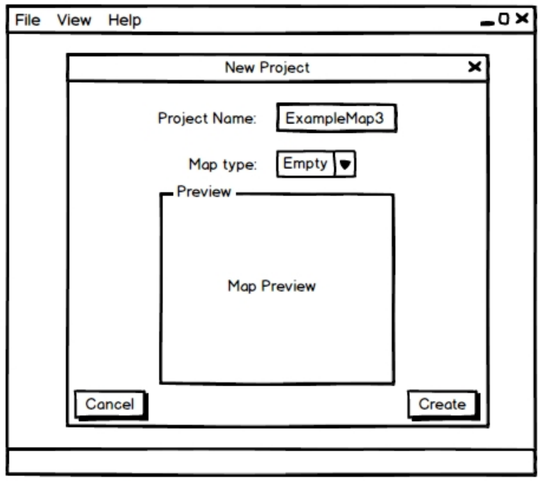
Create Project / Map



[File is clicked]



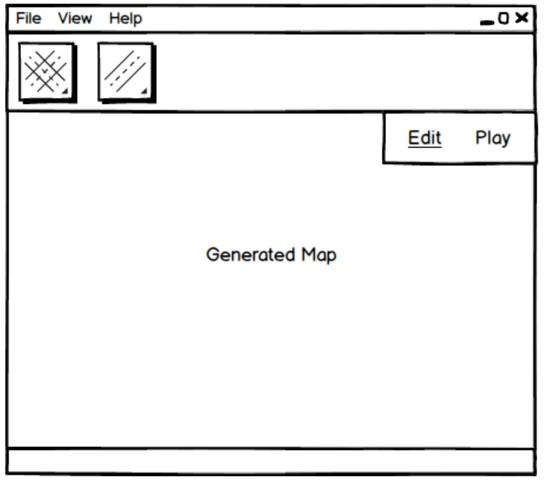
[New is clicked]



Map type can be either:

- empty,
- existing (real-world infrastructure, but with only 4-way stop lights),
- pre-defined (students can download traffic infrastructure files that have been created by someone else, i.e. the teacher)

[A new project is created (this is how you create each map]



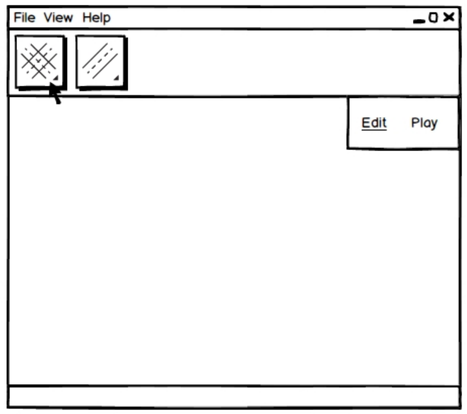
Adding Intersections / Roads to Map

Add Intersection Add Road

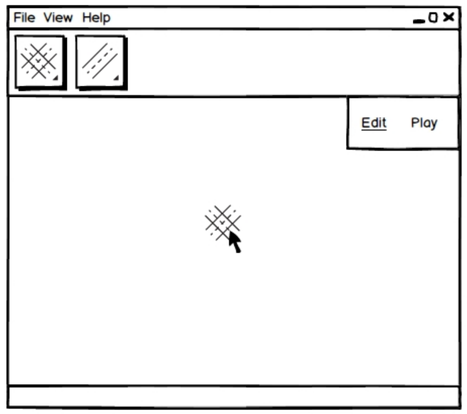
Add Object (by dragging from toolbar)

Note

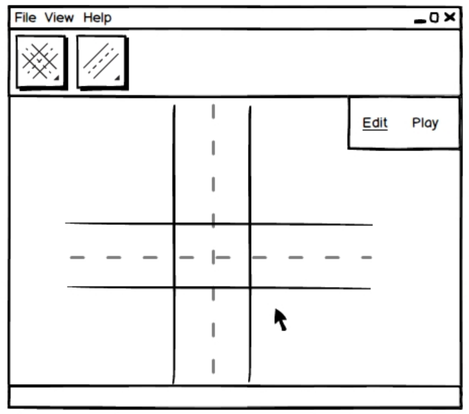
- If one of the Add Object buttons is held for a small amount of time, it will display a window that allows the user to select a different intersection / road that can be used as the default option.



[An Intersection is dragged from the toolbar]

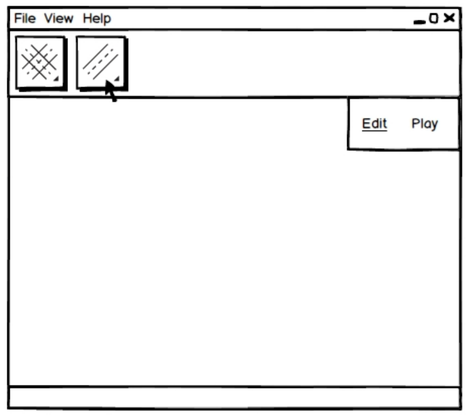


[The intersection is placed on the map.]

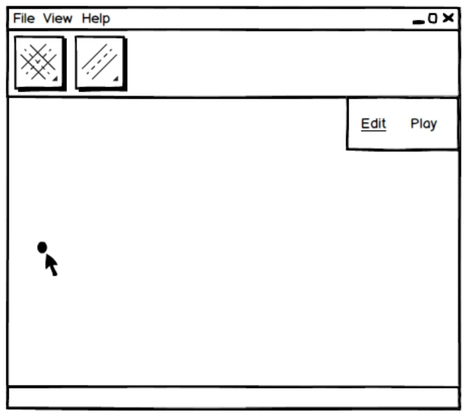


* Notable Scenario: Adding a Road Along a Path

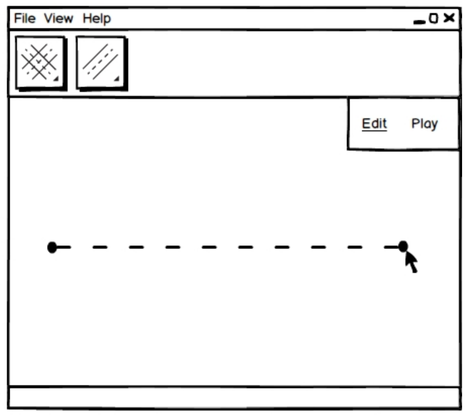
- Besides adding a road by dragging it to the map, you can on also click on the button and then designate a starting and endoing point on the map. This will create a road along the specified path.



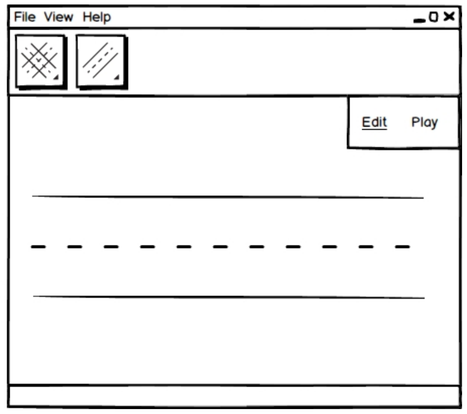
[The Add Road button is clicked. and the starting point graphic appears near the cursor.]



[The starting point icon is specified and the road's path is drawn to the location of the cursor.]



[The end point is specified, which creates the road along the given path.]



* Note about Connecting Roads to an Intersection

- Roads can be connected to an intersection by dragging one end point of a road to the corresponding side of the intersection.
- If the road doesn't have the same number of lanes as the intersection that it is connected to, its lanes will automatically merge or bridge out in order to be compatible with the intersection.

Crash Scenario

Scenario - Overview

The user sets up this intersection with the following condition:

Condition (let's call it Condition1)

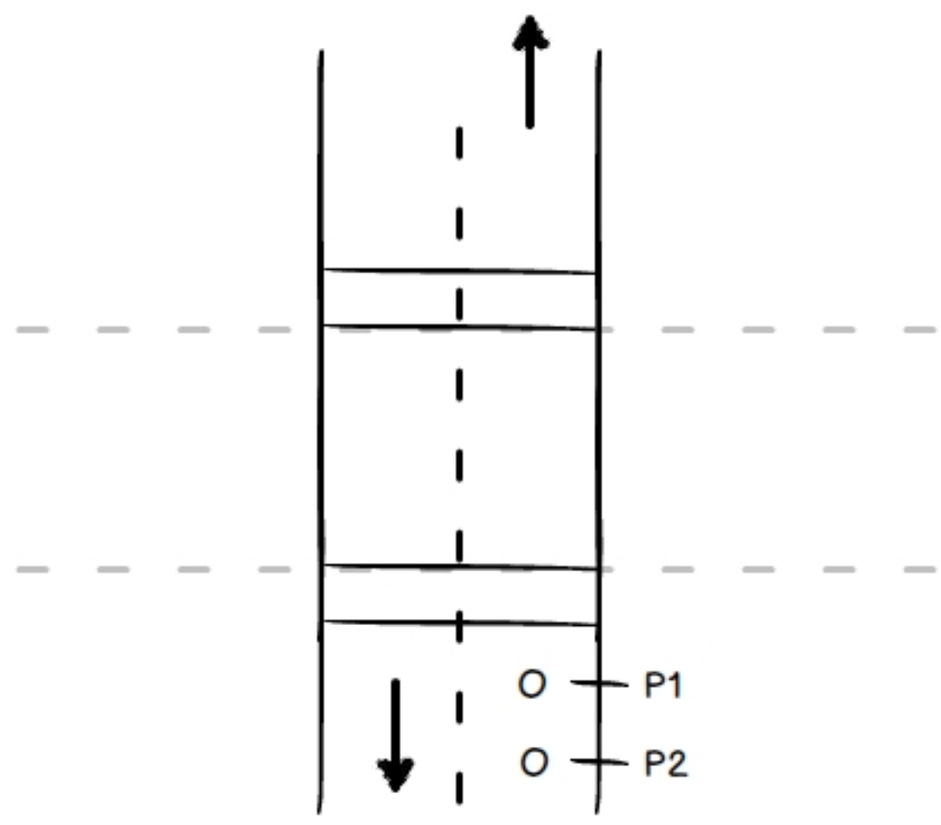
if P1 & P2

Sense Pressure

then Increase timer of

Opposing Traffic Light Pair

by 10 minutes



Critical Note: The default timer for both traffic light pairs was 2 minutes

Reason for possible collisions:

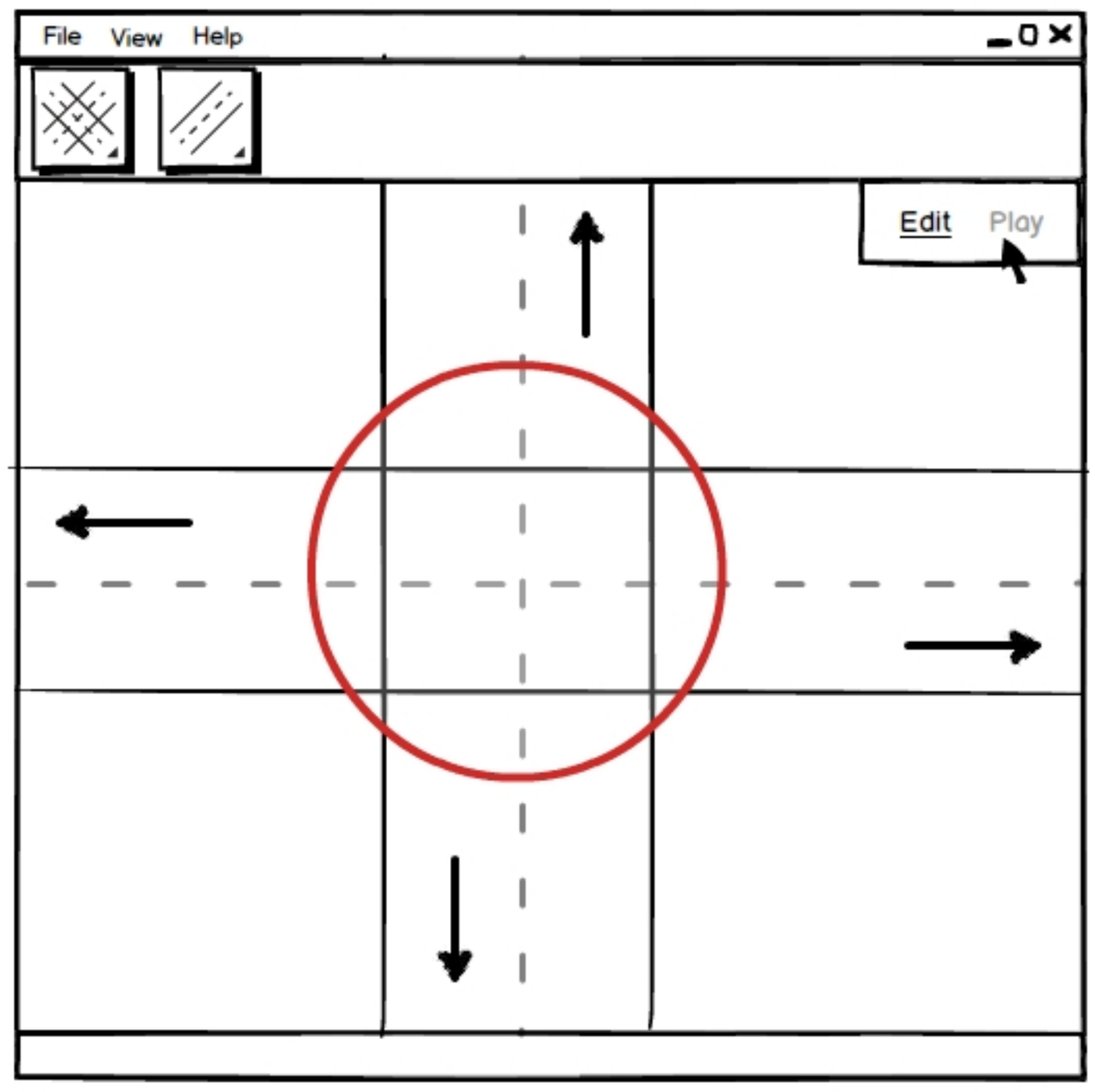
When the timers for the North/South traffic light pair finishes,

then the timer for the West/East traffic light pair may still be going on,

which will produce a collision.

Example of what the user will see

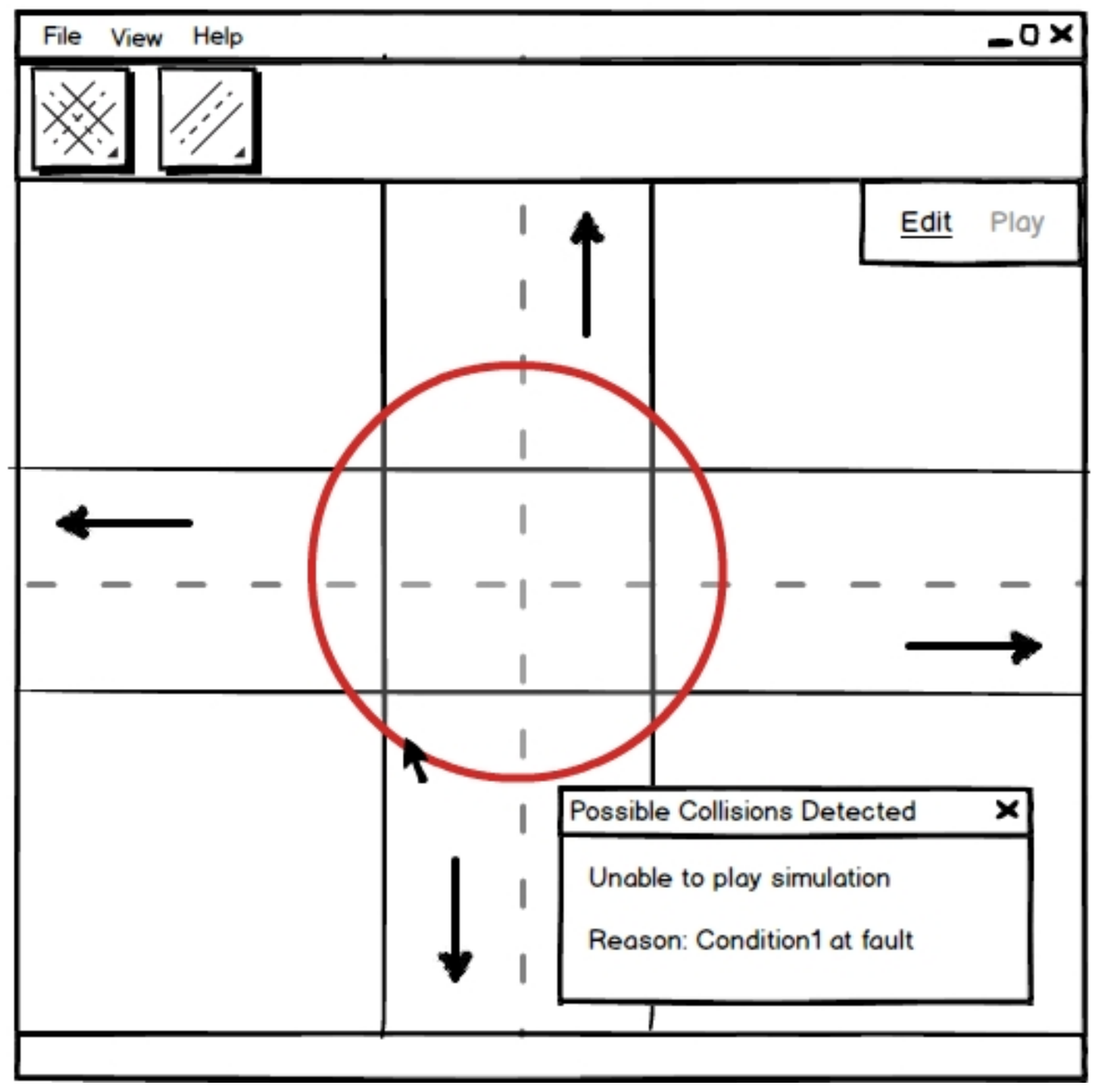
- The user won't be able to click play because its button will be grayed out.



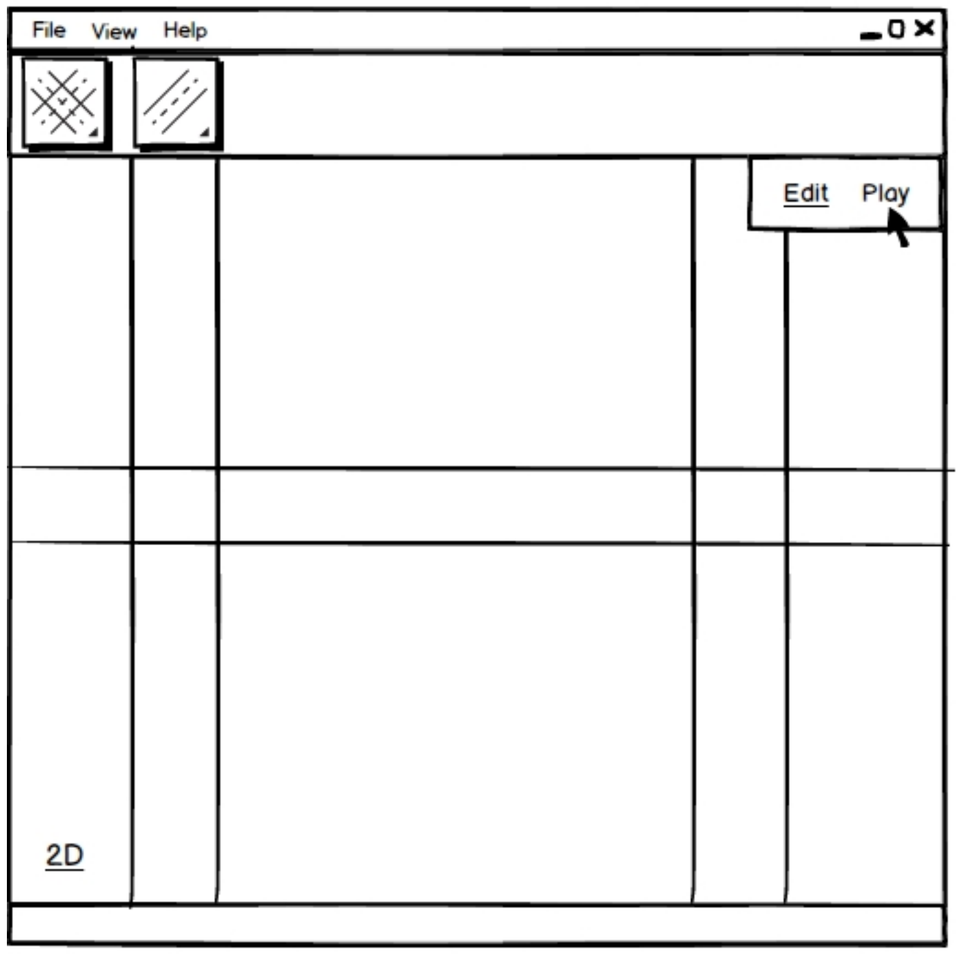
[The user clicks on the intersection.]

- The user will be presented with a dialog box that

specifies the error when the user clicks on the intersection in question.

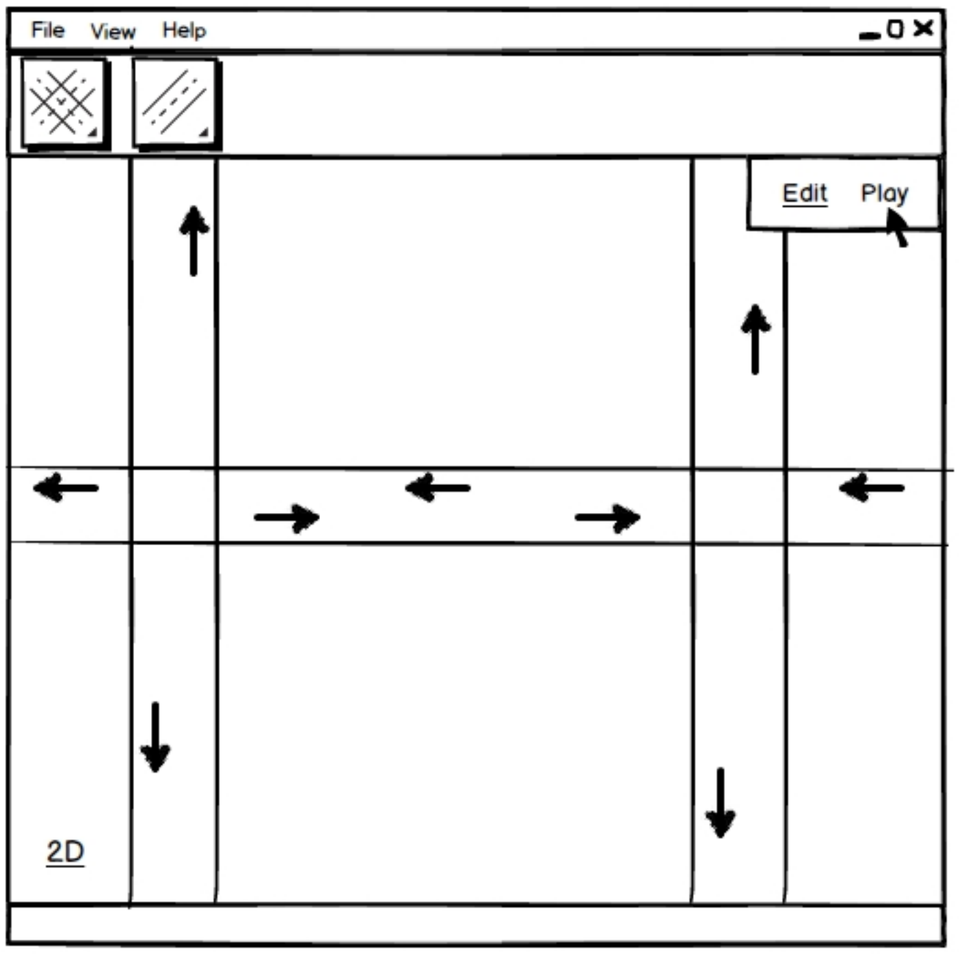


Traffic Simulation

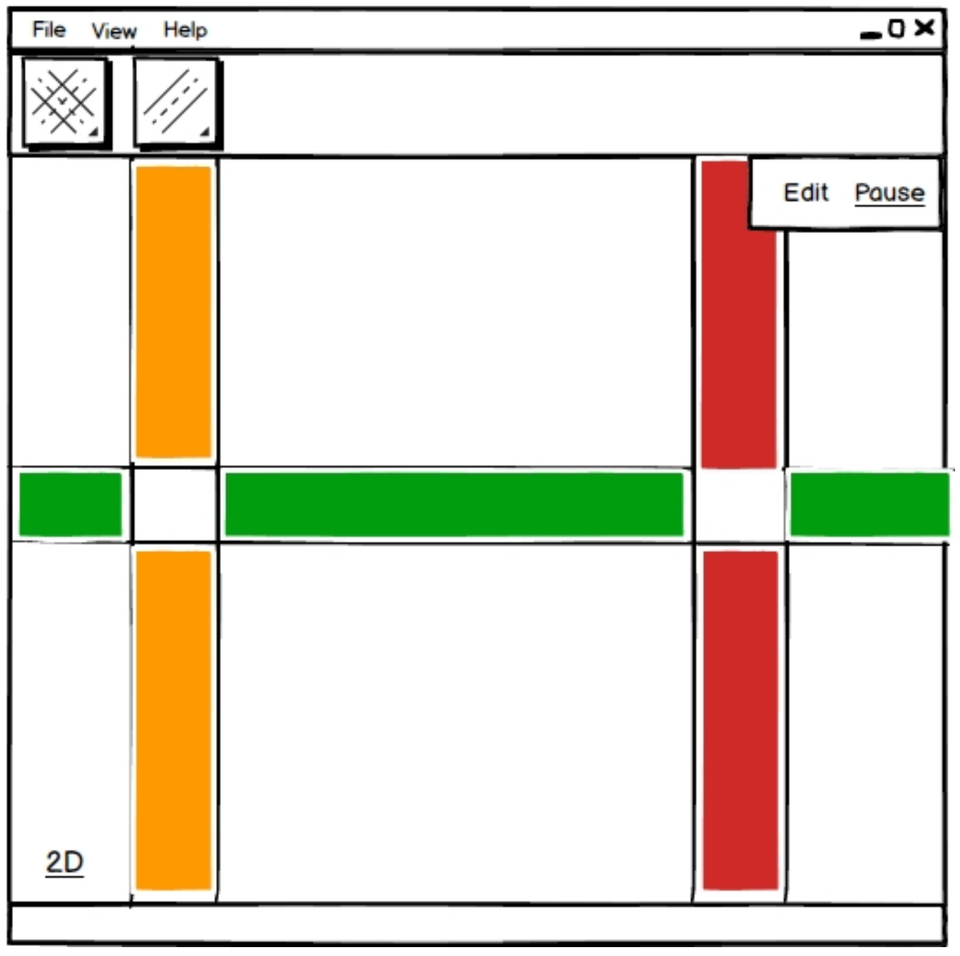


[Play Button is clicked]

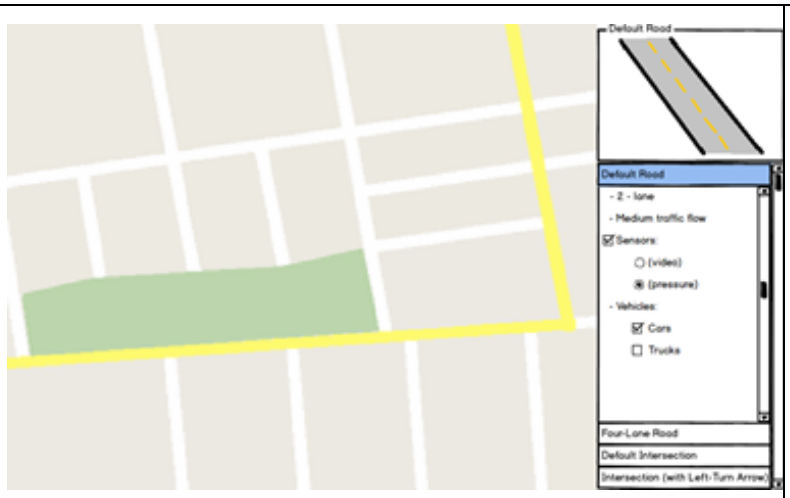
(Play changes to Pause, which can be clicked to toggle between Play and Pause)



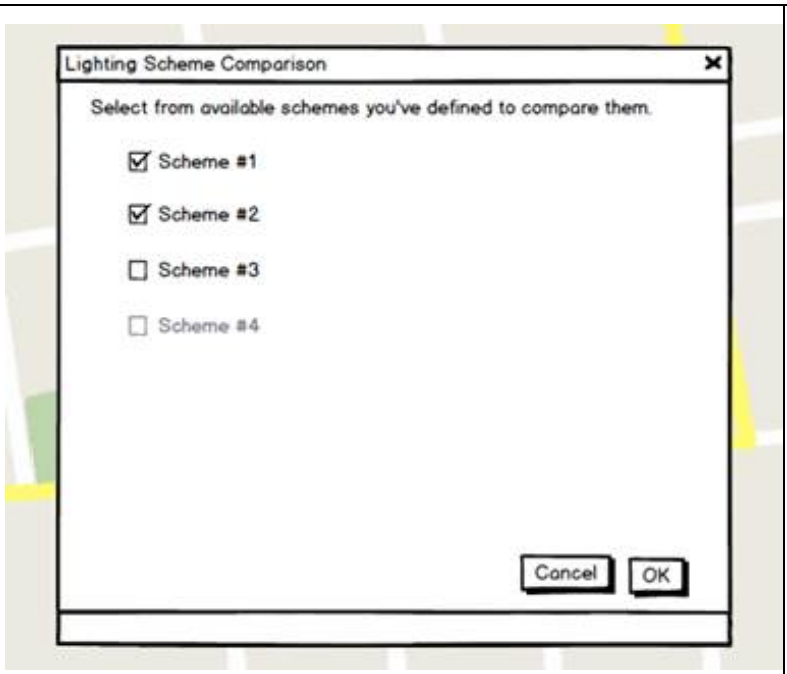
[When traffic flow is shown]



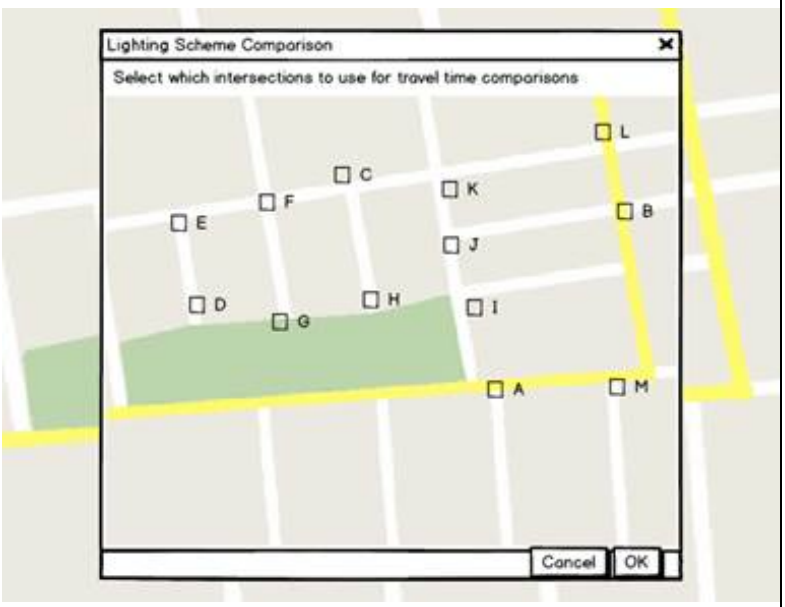
Define Attributes for Individual Roads:

	Default Road: - For ease of use, a 2-lane road is assigned as the default road value. - The user is then able to redefine, the road as they see fit, if they want other options (i.e. sensors, what kind of cars will be used, etc…) for that road.
	A Four-Lane Road: - The user is able to redefine, the road as they see fit, if they want other options (i.e. sensors, what kind of cars will use the road, etc…) for that road.

Lighting Scheme Comparison Tool:

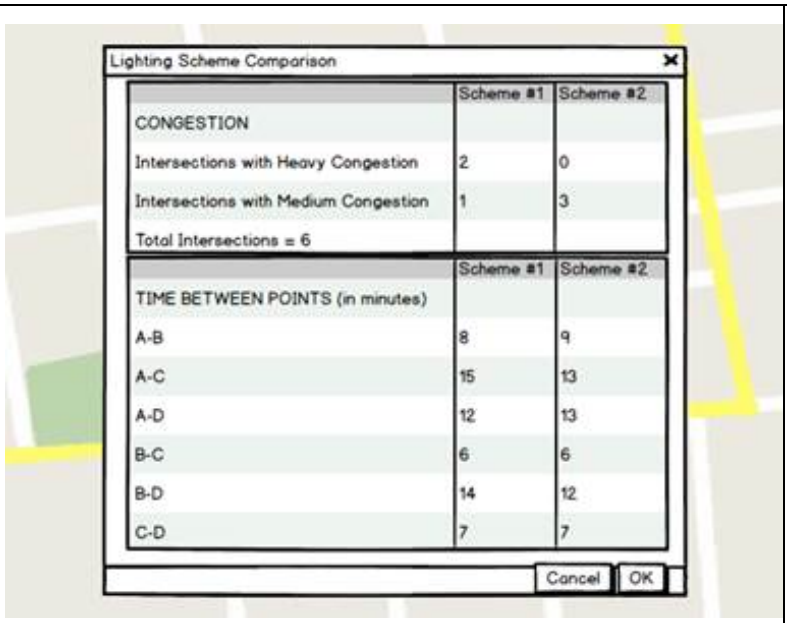
<table>
<tr>
<td> 
</td>
<td><u>Pick from saved schemes to compare:</u>

-Displays all saved lighting schemes for a given map, then allows the user to select which ones they wish to compare.

- Lighting Scheme 4 is not selectable because it contains an invalid lighting scheme leading to a crash scenario.</td>
</tr>
<tr>
<td> 
</td>
<td><u>Select Travel Points:</u>

- Displays the map with all of the intersections as selectable fields.

- Prompts the user to select intersections for the program to use as comparison points to show the relative times it will take to get between these points in the final comparison between schemes.</td>
</tr>
</table>



Final Comparison Details:

- Shows the final side-by-side comparison of the selected lighting schemes.

- Compares by congestion showing the number of intersections which cause heavy congestion, and medium congestion.

- Also shows the time it would take to move between the user selected intersections.